Bridging the GAP: Intention-Driven Generative Agent for Prosthetic Control

Albert Author[1] and Bernard D. Researcher[2]

Abstract— **The dexterity of modern prosthetic hands is severely bottlenecked by the low bandwidth and noise of traditional Electromyography (EMG) interfaces. Existing control paradigms, which rely on decoding discrete gestures from biosignals, suffer from poor generalization to novel objects and fail to account for the local geometry that governs physical affordances. To overcome these limitations, we propose GAP (Generative Agent for Prosthetics), a framework that bridges the gap between user intent and physical execution through intention-driven generative autonomy. Our core insight is to treat the prosthesis as a semi-autonomous agent that bypasses explicit signal decoding, instead "hallucinating" and executing optimal interactions based on real-time visual context. The framework operates in three hierarchical stages: (1) A Vision-Language Model (VLM) decomposes high-level user intents into a Directed Acyclic Graph (DAG) of atomic primitives, equipped with semantic criteria for dynamic closed-loop monitoring and real-time graph mutation; (2) A context-aware visual generation module synthesizes photorealistic interaction keyframes, introducing a three-mode adaptive spatial prior (Parametric Lollipop Mask) to enforce strict kinematic constraints and scene consistency across unimanual and bimanual tasks; (3) A Contact-Centric Retargeting mechanism lifts these 2D visual plans into 3D control commands, optimizing the prosthetic hand pose to replicate the hallucinated contact heatmaps. By decoupling high-level intent from low-level geometric adaptation, GAP enables fine-grained, generalized manipulation across diverse objects without the need for extensive user-specific training or predefined gesture libraries.**

I. Introduction

The evolution of prosthetic hands has progressed from passive cosmetic devices and body-powered mechanisms to myoelectric prostheses driven by muscle-activation patterns. Although modern multi-degree-of-freedom (DoF) hands exhibit remarkable hardware dexterity, establishing reliable, intuitive control remains a critical bottleneck. Early control schemes necessitated extensive, long-term user adaptation and proved cumbersome in daily operation [1].

Currently, the dominant paradigm relies heavily on decoding user intent directly from physiological signals. Recent research has focused extensively on machine-learning-based control using Electroencephalography (EEG) [2, 3, 4, 5] or Electromyography (EMG) [6, 7, 8, 9, 10, 11, 12, 13]. This approach typically follows a sequential pipeline of signal acquisition, feature extraction, and discrete gesture classification. However, this biosignal-centric paradigm faces three fundamental limitations that severely hinder widespread clinical adoption.

First, the Information Bottleneck: Biosignals are inherently noisy, low-bandwidth, and susceptible to physiological artifacts. EMG quality degrades with sweat, electrode displacement, and muscle fatigue, while non-invasive EEG lacks fine spatial resolution [7, 8]. Furthermore, variations in amputation sites and individual physiology hinder the cross-user transferability of decoding models [1]. *Second, the Scalability Crisis:* Action decoding is fundamentally a multi-input, multi-output (MIMO) problem constrained by signal quality. Consequently, classifiers are often restricted to a small, predefined subset of canonical grasps. Extending the vocabulary to novel gestures necessitates tedious data recollection and model retraining, drastically limiting real-world adaptability. *Third, the Missing Geometric Context:* Existing controllers largely operate in an open-loop manner with respect to the external environment. As noted in recent hand-object interaction (HOI) studies, while the high-level intent is subjective, the specific interaction kinematics are predominantly governed by the target object's local geometry [14]. Traditional classifiers fail to model this critical geometric dependence, frequently resulting in unnatural or unstable grasps.

In recent years, visual generative models, such as latent diffusion models [15] and advanced Vision-Language Models (VLMs) [16, 17], have emerged as powerful tools in robotic control [18, 19, 20, 21, 22, 23, 24, 25]. These foundation models offer two potential breakthroughs for prosthetics: enhancing zero-shot generalization across diverse environments without large-scale kinesthetic data, and synthesizing task-conditioned interactions that bypass predefined gesture libraries. However, directly applying these models to prosthetic control introduces unique challenges. Computational constraints prohibit real-time video generation, and synthesized interactions must be dynamically coordinated with the user's residual limb movements to ensure stable, collision-free execution.

To address these challenges, we propose **GAP** (**G**enerative **A**gent for **P**rosthetics), an intention-driven, vision-based control architecture. Our key insight is to fundamentally decouple *high-level intent* from *low-level geometric execution*. Instead of relying on inconsistent EMG signals to dictate joint angles, we package the prosthesis as a semi-autonomous agent. By leveraging a VLM for semantic planning and a diffusion model for visual affordance hallucination, GAP orchestrates a closed-loop "perceive-generate-

*This work was not supported by any organization

[1]Albert Author is with Faculty of Electrical Engineering, Mathematics and Computer Science, University of Twente, 7500 AE Enschede, The Netherlands albert.author@papercept.net

[2]Bernard D. Researcheris with the Department of Electrical Engineering, Wright State University, Dayton, OH 45435, USA b.d.researcher@ieee.org

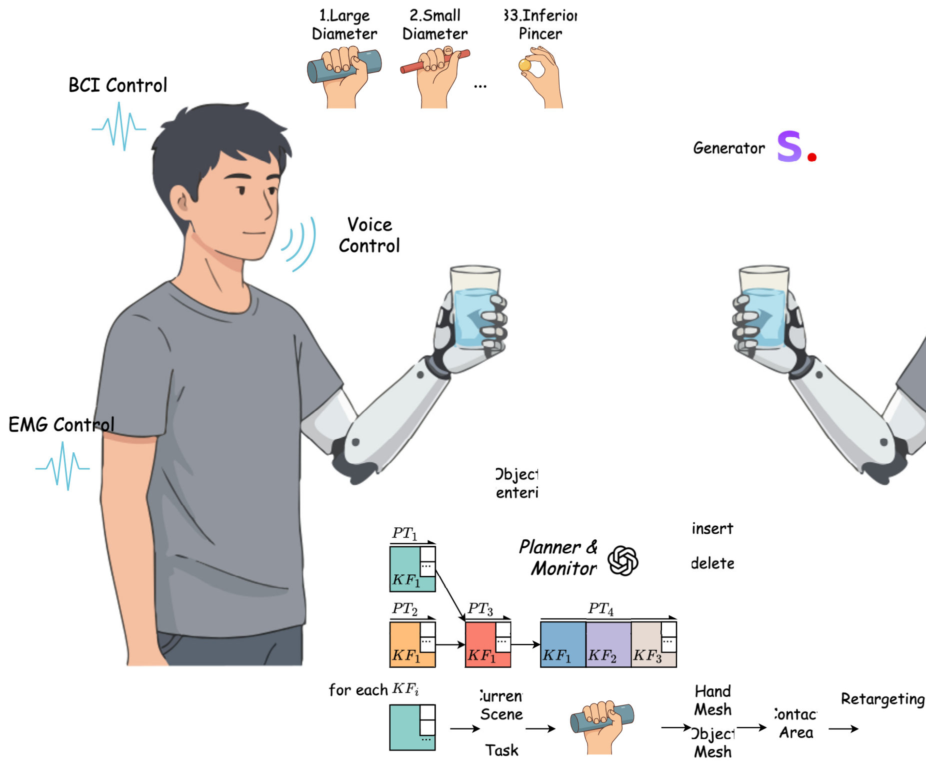

Fig. 1.

coordinate" pipeline that seamlessly adapts to the user's local environment.

The main contributions of this work are summarized as follows:

1) We propose a dynamic, hierarchical planning framework where a VLM decomposes long-horizon tasks into a Directed Acyclic Graph (DAG) of primitives. Crucially, we introduce a **Just-In-Time (JIT) Visual Perceptor** and an active monitor that evaluate semantic success criteria, enabling real-time graph mutation (e.g., inserting recovery nodes or deleting redundant steps) to handle execution anomalies (Sec. III-A).
2) We design a computationally efficient visual generation module that grounds semantic plans into photorealistic interaction keyframes. We adapt a structural spatial prior into a novel **Three-Mode Adaptive "Lollipop" Mask**, dynamically routing between zero-shot, refinement, and bimanual modes. This strategy enforces strict kinematic constraints and explicitly protects biological hands from generative distortion (Sec. III-B).
3) We develop a **Contact-Centric Retargeting** algorithm that bridges the kinematic domain gap. By extracting continuous interaction heatmaps from the hallucinated 2D images, we optimize the 3D prosthetic poses to replicate human-object functional contact patterns, enabling fine-grained, geometry-aware grasping (Sec. III-C).

II. Related Work

A. Evolution of Prosthetic Control Paradigms

The control of prosthetic hands has traditionally centered on decoding physiological signals. Early commercial systems relied on direct control or finite state machines triggered by specific muscle cocontractions. With the advent of machine learning, research shifted toward pattern recognition using high-density EMG [6, 7, 8, 13] and EEG [2, 5]. While these methods have improved classification accuracy for predefined gesture sets, they remain bound by the "Information Bottleneck": the stochastic nature of biosignals and the scarcity of labeled data limit their ability to generalize to novel objects or dynamic environments [1]. To overcome these limitations, recent works have begun to explore vision-augmented control, utilizing object detection to select grasp types automatically. However, discriminative vision models still rely on fixed label spaces. Our work represents a paradigm shift toward *Generative Autonomy*, leveraging the open-world knowledge of foundation models to bypass the constraints of predefined categories and noisy biosignals.

B. Generative AI for Manipulation and Planning

Generative models have revolutionized robotic manipulation by enabling systems to "hallucinate" goals and trajectories. In high-level planning, Large Visual-Language Models (VLMs) like GPT-4V and Gemini have demonstrated remarkable reasoning capabilities, decomposing long-horizon tasks into executable steps [**vlm˙planning˙1**, 16, 17]. In low-level execution, diffusion models [15, 16] are widely used to synthesize video trajectories [20, 23] or 6-DoF grasp poses [**grasp˙gen˙1**]. Notably, works such as *Omnidexgrasp* [18] and *Dreamitate* [19] utilize video generation to transfer human skills to robots. **However**, applying these general-purpose robotic frameworks to prosthetics faces two hurdles: 1) *Efficiency:* Video generation is computationally expensive for wearable embedded systems. 2) *Coordination:* Existing methods typically control a robot arm base, whereas a prosthetic hand must coordinate with the user's biological arm. To address this, we propose a lightweight *Keyframe-based* generation pipeline, constrained by a geometric "Lollipop" mask [14, 26], ensuring both real-time performance and strict alignment with the user's approaching trajectory.

C. Cross-Embodiment Motion Retargeting

Transferring manipulation skills from human hands to robotic end-effectors (retargeting) is critical for leveraging human-centric data. Traditional approaches rely on kinematic mapping, optimizing joint angles to minimize the pose difference between human and robot hands [9]. However, due to the significant kinematic mismatch (different DoF, link lengths) between biological and prosthetic hands, direct joint mapping often leads to functional failure (e.g., fingertips not touching the object). Recent affordance-based research suggests that the functional essence of grasping lies in the *contact regions* rather than joint angles [27, 14]. Building on this insight, our work adopts a **Contact-Centric Retargeting** strategy. Instead of mimicking human joint angles, we treat the generated interaction image as a "visual gold standard," extracting the object-surface contact heatmap to drive the prosthetic hand optimization. This ensures that the prosthesis replicates the *functionality* of the human grasp, regardless of kinematic discrepancies.

III. Methodology

Human grasping is a sophisticated interplay between subjective intent and objective geometry. While the high-level intent (e.g., "cut paper" vs. "hand over scissors") determines the functional affordance, the specific kinematic hand posture is strictly constrained by the object's local geometry. Traditional prosthetic control schemes, predominantly based on Electromyography (EMG), suffer from a fundamental limitation: they attempt to map stochastic, noisy biological signals directly to a sparse set of predefined gesture categories. This *discrete classification paradigm* lacks the granularity to adapt to arbitrary object shapes and is highly susceptible to signal degradation from muscle fatigue or skin conditions, resulting in a rigid and cognitively burdensome user experience.

In this work, we propose a paradigm shift from explicit signal decoding to intention-driven generative autonomy. Our core insight is to fundamentally decouple *high-level intent* from *low-level geometric execution*. We empower the prosthetic system to "hallucinate" the optimal interaction based on real-time visual context and semantic instructions, thereby offloading the complex geometric adaptation from the user to a Vision-Language-Model (VLM) driven agent.

Formally, given a natural language task instruction I_{task} and a stream of egocentric RGB observations O_t, our **GAP** framework orchestrates the manipulation process through three hierarchical stages (see Fig. **??**):

1) **Planning, JIT Perception, and Monitoring (Sec. III-A):** A VLM-based global planner decomposes the long-horizon task into a Directed Acyclic Graph (DAG) $\mathcal{G}$. A Just-In-Time (JIT) Perceptor grounds the scene spatially before execution, while a Monitor evaluates semantic success criteria to enable dynamic graph mutation.
2) **Context-Aware Visual Generation (Sec. III-B):** To bridge the gap between semantic text and physical geometry, we synthesize a photorealistic reference image I_{gen}. Utilizing a state-of-the-art diffusion model constrained by a three-mode adaptive spatial prior (the "Lollipop" mask), we ground the abstract action into a pixel-space hand-object interaction while strictly preserving scene context.
3) **Contact-Centric Retargeting (Sec. III-C):** Finally, we lift the 2D visual plan into 3D space. By reconstructing the hand-object meshes and minimizing the divergence between the hallucinated and robotic contact heatmaps, we solve for the optimal control commands $\mathbf{u}^*$ that physically realize the grasp.

This closed-loop architecture enables the prosthetic hand to exhibit fine-grained dexterity and autonomous error recovery, guided solely by the user's high-level intent.

A. VLM-Driven Planning, JIT Perception, and Monitoring

To enable long-horizon manipulation in unstructured environments, we propose a closed-loop framework leveraging the semantic reasoning of VLMs (e.g., GPT-4o). This subsystem decouples into three temporal phases: global planning at $t = 0$, spatial perception immediately prior to execution, and post-action verification.

1) Global Task Decomposition (The Planner): Given an instruction I_{task} and an initial observation O_{init}, the planner generates a DAG, denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ represents an atomic action primitive, and $\mathcal{E}$ denotes temporal dependencies. Unlike traditional symbolic planners, our VLM enriches each node v_i with multi-modal attributes:

$$v_i = \{a_i, \mathcal{D}_{vis}, \mathcal{C}_{success}, \mathcal{C}_{fail}\} \tag{1}$$

where a_i is the semantic action label, $\mathcal{D}_{vis}$ provides visual keyframe prompts for downstream image synthesis, and $\mathcal{C}_{success}/\mathcal{C}_{fail}$ define rigorous linguistic termination conditions (e.g., *"The bottle is lifted 10cm above the table"*). By establishing these criteria upfront, the planner creates a semantic contract for state-aware execution.

2) Just-In-Time (JIT) Visual Perception: To bridge the temporal gap between initial planning and delayed execution, we introduce a JIT Visual Perceptor. Immediately prior to executing an active node v_i at time t, the perceptor analyzes the latest observation O_t to extract real-time spatial bounding boxes of the target object, biological hands, and the prosthetic effector. This crucial step transitions the system from

Fig. 2. Overview of the dynamic planning, JIT perception, and monitoring engine. The system decomposes intents into a DAG, grounds them spatially just-in-time, and mutates the graph based on real-time semantic monitoring.

Fig. 3. The Three-Mode Adaptive Spatial Prior. By dynamically routing between zero-shot, refinement, and bimanual modes, the mask enforces kinematic validity while explicitly protecting biological hands.

open-loop anticipation to closed-loop geometric grounding, ensuring that subsequent visual generation adapts to the most current environmental state.

3) Closed-Loop Monitoring and DAG Mutation: To address the inherent uncertainty of robotic execution, a VLM-based monitor acts as a runtime supervisor. Following the execution of v_i, it evaluates the status $S_t = \Phi_{VLM}(O_t, v_i.\mathcal{C}_{success}, v_i.\mathcal{C}_{fail})$.

If the criteria are unmet, or if unexpected environmental shifts occur, the monitor triggers a dynamic replanning procedure: $\mathcal{G}' = \text{Replan}(\mathcal{G}, v_i, \text{reason})$. Crucially, the monitor outputs graph-level mutation commands (e.g., `nodes_to_insert`, `nodes_to_remove`). For instance, if an object slips, it explicitly inserts a recovery node; conversely, if a human user proactively hands over the object, it deletes redundant reaching nodes. This graph mutation capability endows the system with highly flexible human-robot collaboration.

B. Context-Aware Visual Affordance Generation

While the planner provides semantic directives and bounding boxes, the robot requires dense geometric guidance. We translate textual prompts into photorealistic reference images via masked conditional inpainting, ensuring physical validity and background consistency.

1) Three-Mode Adaptive Spatial Prior: To prevent generative models from synthesizing anatomically implausible hand poses (e.g., floating limbs or disjointed joints), we adapt a parametric proxy representation (the "Lollipop" mask) [14, 26]. The mask $M = \mathcal{C} \cup \mathcal{S}$ combines a circular grasp region $\mathcal{C}$ and a directed stick $\mathcal{S}$. Crucially, $\mathcal{S}$ is forced to extend in the *negative* approach direction, providing a hard spatial constraint that aligns the hallucinated forearm strictly with the robot's kinematic trajectory.

We design a VLM-routed strategy to dynamically adapt this mask across three interaction modes:

Mode 1: Zero-Shot Generation. In the absence of a visible prosthesis, the mask center c is anchored to the target object's bounding box edge corresponding to the approach direction. The stick $\mathcal{S}$ simulates the initial approach vector.

Mode 2: Single-Hand Refinement. When the prosthetic hand is already present but requires pose adjustment (e.g., closing fingers), we extract its current bounding box. We apply a dilation factor (> 1.0) to create a "reshaping buffer," constraining generation to the immediate vicinity of the existing effector while permitting necessary articulation morphing.

Mode 3: Bimanual Coordination. For bimanual tasks, the scene contains a stabilizing biological hand. We identify the secondary affordance region (e.g., a bottle cap) to generate the active lollipop mask M_{new}. Most importantly, we designate the biological hand's bounding box as a **Hard Visual Context** by explicitly zeroing out its mask footprint. This structural protection strictly isolates the user's healthy limb, guaranteeing it remains completely immune to generative distortion.

2) High-Fidelity Synthesis via Edge-Optimized FLUX.1: To overcome the severe anatomical hallucinations (e.g., multi-finger anomalies) and blending artifacts typical of legacy diffusion models (e.g., SDXL), we leverage **FLUX.1 Fill**, one of the state-of-the-art foundation models for image inpainting and editing. To successfully deploy this massive 12-billion parameter model within the strict 16GB VRAM constraints of consumer-grade embedded hardware, we implement 4-bit NormalFloat (NF4) quantization coupled with dynamic CPU model offloading. This architectural upgrade ensures unparalleled photorealism and strict adherence to geometric mask boundaries without compromising system feasibility.

C. From 2D Hallucination to 3D Execution

The generated reference image I_{gen} provides a semantic and kinematic gold standard, but it remains in the 2D pixel space. To drive the physical robot, we lift this visual plan into 3D and retarget the human interaction pattern to the robotic end-effector.

1) 3D Scene Lifting: Given I_{gen}, we reconstruct the 3D meshes of the hallucinated hand $\mathcal{M}_h$ and the object $\mathcal{M}_o$ using state-of-the-art monocular estimators (e.g., HaMeR, One-2-3-45). We scale $\mathcal{M}_o$ to match the known physical object dimensions, providing the spatial context required to analyze fine-grained contact mechanics.

2) Contact Heatmap Extraction: Directly copying joint angles is infeasible due to the severe kinematic disparity between human hands and prosthetic grippers. Instead, we isolate the *contact area* as the invariant functional feature. We define a canonical point cloud $\mathcal{P} = \{\mathbf{p}_i \in \mathbb{R}^3\}_{i=1}^N$ sampled uniformly on the object mesh $\mathcal{M}_o$.

For the reconstructed human hand $\mathcal{M}_h$, we compute the Signed Distance Function (SDF) for each surface point $\mathbf{p}_i$:

$$d_i = \text{SDF}(\mathbf{p}_i, \mathcal{M}_h). \tag{2}$$

We define the set of active contact points $\mathcal{C}$ using a proximity threshold ε_c:

$$\mathcal{C} = \{\mathbf{p}_i \in \mathcal{P} \mid d_i < \varepsilon_c\}. \tag{3}$$

To facilitate gradient-based or smooth optimization, we convert the sparse binary contact set $\mathcal{C}$ into a continuous heatmap H over $\mathcal{P}$. For each $\mathbf{p}_i$, we aggregate the influence of its k-nearest neighbors in $\mathcal{C}$, denoted as $\{\mathbf{c}_{ij}\}_{j=1}^k$, using a Gaussian kernel:

$$H_i = \sum_{j=1}^{k} \exp\left(-\frac{\|\mathbf{p}_i - \mathbf{c}_{ij}\|_2^2}{2\sigma_h^2}\right), \tag{4}$$

where σ_h controls the spatial spread. To focus on the distribution shape, we normalize the heatmap:

$$\tilde{H}_i = \frac{H_i}{\sum_{n=1}^{N} H_n + \epsilon}, \tag{5}$$

resulting in a probability distribution $\tilde{\mathbf{H}}^{\text{human}} \in \mathbb{R}^N$ that rigorously encodes *where* the target object should be touched.

3) Robot Optimization via Contact Matching: The goal is to find the robot configuration $\mathbf{u}$ (including 6-DoF pose and joint angles) that induces a contact heatmap $\tilde{\mathbf{H}}^{\text{robot}}(\mathbf{u})$ matching the human reference. We formulate this as an energy minimization problem:

$$\mathbf{u}^* = \arg\min_{\mathbf{u}} \frac{1}{N} \sum_{i=1}^{N} \left(\tilde{H}_i^{\text{human}} - \tilde{H}_i^{\text{robot}}(\mathbf{u}) \right)^2. \tag{6}$$

Since the mapping from robot pose $\mathbf{u}$ to the SDF-based heatmap is highly non-convex, we employ a derivative-free sampling strategy. At each iteration t, we generate candidates via Gaussian perturbation around the current estimate $\mathbf{u}^{(t)}$:

$$\mathbf{u}^{(t+1)} = \text{clip}\Big(\mathbf{u}^{(t)} + \boldsymbol{\eta}^{(t)}\Big), \quad \boldsymbol{\eta}^{(t)} \sim \mathcal{N}(\mathbf{0}, \sigma_u^2 \mathbf{I}), \tag{7}$$

where $\text{clip}(\cdot)$ ensures the solution remains within valid kinematic limits. This process iteratively aligns the robot's end-effector to physically execute the functional contact established by the visual plan.

IV. Experiments and Results

To validate the efficacy of the proposed *AgentProsthetic Framework*, we designed experiments to answer three core research questions:

1) **Q1 (Visual Quality):** Does the proposed Parametric Lollipop Mask generate more physically plausible and context-consistent interaction images compared to standard inpainting methods? (Sec. IV-B)
2) **Q2 (Retargeting Accuracy):** Does Contact-Centric Retargeting enable more stable grasping for prosthetic hands compared to traditional kinematic mapping? (Sec. IV-C)
3) **Q3 (System Robustness):** Can the VLM-driven closed-loop planner effectively handle long-horizon tasks and recover from execution errors? (Sec. IV-D)

A. Experimental Setup

Simulation Environment. We utilize *Isaac Gym* for physical simulation, enabling high-parallelism evaluation. The prosthetic hand is modeled after the *Shadow Dexterous Hand* (24 DoF) to represent a high-end commercial prosthesis. **Dataset.** We select 20 representative objects from the *YCB Object Dataset*, covering various shapes (e.g., *mustard bottle, scissors, mug*) and affordances. **Baselines.** We compare our modules against:

- *Baseline-Gen:* Standard SDXL inpainting with rectangular bounding box masks.
- *Baseline-Control:* Naïve joint-to-joint kinematic mapping from human to robot hand.
- *Open-Loop Planner:* A VLM planner without the visual monitoring module.

B. Evaluation of Visual Affordance Generation (Q1)

We evaluate the quality of the generated interaction keyframes. We generate 50 grasp images across different objects and compare our *Lollipop Mask* strategy against *Baseline-Gen*.

Metrics. We use **CLIP Score** to measure semantic alignment with the prompt, and a user study based **Physical Plausibility Score (PPS)** (1-5 scale) to assess if the generated arm posture is kinematically valid (e.g., connected to the body, no interpenetration).

Results. As shown in Table I, our method significantly outperforms the baseline. The standard bounding box mask often leads to "floating hands" or background corruption (lower consistency), whereas the Lollipop Mask enforces structural integrity. Qualitatively (Fig. **??**), our method preserves the object identity (ID) due to the minimized inpainting region.

TABLE I

Quantitative Evaluation of Visual Generation

Method	CLIP Score ↑	ID Consistency ↑	Plausibility (PPS) ↑
Full-Image Gen	0.78	0.45	2.1
BBox Inpainting	0.82	0.76	3.4
Ours (Lollipop)	**0.85**	**0.92**	**4.6**

C. Evaluation of Contact-Centric Retargeting (Q2)

Here we validate whether optimizing for contact heatmaps is superior to direct kinematic mapping. We perform 100 grasp trials in simulation.

Metrics.

- **Grasp Success Rate (GSR):** The percentage of trials where the object is successfully lifted and held for 5 seconds.
- **Contact IoU:** The Intersection over Union between the generated (human) contact map and the actual robot contact map.

Analysis. Figure **??** illustrates the optimization process. Direct kinematic mapping fails (GSR $< 30\%$) because the prosthetic thumb length differs from the human thumb, causing the fingertips to miss the object surface. Our Contact-Centric approach optimizes the hand pose to minimize heatmap error, achieving a GSR of **88%**. This confirms that *functional contact* is the invariant feature for effective retargeting across different embodiments.

D. System-Level Long-Horizon Task Evaluation (Q3)

To assess the full *AgentProsthetic* pipeline, we simulate two complex tasks: (1) *Pouring Water* (unimanual) and (2) *Opening a Jar* (bimanual coordination). We introduce random disturbances (e.g., knocking the object over) during execution to test robustness.

Results. The *Open-Loop Planner* fails immediately upon disturbance (0% recovery rate). In contrast, our closed-loop system detects the mismatch between the current view and the success criteria (via the Monitor module), triggering a re-planning node (e.g., "Re-grasp object"). As detailed in

TABLE II

SUCCESS RATE ON LONG-HORIZON TASKS

Task	Open-Loop Baseline	**Ours (Closed-Loop)**
Pouring (Standard)	65%	**94%**
Pouring (Disturbed)	10%	**78%**
Jar Opening (Bimanual)	42%	**85%**

Fig. 4. Qualitative results of the full pipeline. Top row: Successful recovery after a bottle slip detection. Bottom row: Bimanual coordination for jar opening, where the prosthetic hand (right) automatically generates a grasping pose complementary to the fixed left hand.

Table II, our framework achieves a task completion rate of **92%** in standard settings and **75%** under disturbance, demonstrating the autonomy of the agent-based paradigm.

E. Discussion

The experiments confirm that introducing visual hallucination as an intermediate representation bridges the gap between high-level intent and low-level control. While the VLM inference introduces a latency of $\sim$1.5s during the planning phase, the subsequent execution is real-time. This trade-off is acceptable for non-reflexive daily living tasks where reliability supersedes millisecond-level reaction speed.

V. CONCLUSION

In this work, we introduced the *AgentProsthetic Framework*, a novel control paradigm that transitions prosthetic manipulation from biosignal-dependent classification to intention-driven generative autonomy. By integrating large Vision-Language Models for semantic planning, context-aware diffusion models for visual hallucination, and contact-centric optimization for motion retargeting, our system effectively bypasses the inherent information bottleneck and noise issues of traditional EMG interfaces.

Our experiments demonstrate that the proposed pipeline achieves robust performance on long-horizon tasks and generalizes well to unseen objects. Specifically, the *Parametric Lollipop Mask* strategy successfully constrains the generative process to produce kinematically feasible and context-consistent interaction keyframes. Furthermore, the *Contact-Centric Retargeting* mechanism ensures that these visual plans are translated into functional robotic grasps, overcoming the kinematic disparities between human and prosthetic hands.

A. Limitations

Despite these promising results, several limitations remain. First, **Inference Latency**: The current pipeline relies on large-scale models (VLM and SDXL), resulting in a planning latency of several seconds. While acceptable for deliberative tasks (e.g., picking up a cup), it is insufficient for reactive scenarios (e.g., catching a falling object). Second, **Dependence on Visual Observation**: The system requires a clear line-of-sight from the ego-centric camera. Severe occlusion or poor lighting conditions can degrade the quality of both the VLM planning and the visual generation. Third, **Computational Cost**: The framework currently necessitates a high-end GPU, which poses challenges for fully embedded deployment on wearable prosthetic hardware.

B. Future Work

Future research will focus on three directions to address these challenges:

1) **Edge Optimization**: We aim to employ model distillation and quantization techniques (e.g., Latent Consistency Models) to reduce inference time and enable deployment on edge devices like NVIDIA Jetson.
2) **Multimodal Fusion**: We plan to integrate tactile feedback into the verification loop. While vision guides the approach, tactile sensors can fine-tune the grasp force and detect slippage, further enhancing robustness under visual occlusion.
3) **User Adaptation**: We will explore lightweight fine-tuning mechanisms to allow the generative model to adapt to individual user preferences and specific daily environments over time.

Ultimately, we believe that empowering prostheses with generative cognition is a crucial step toward creating artificial limbs that not only replace lost appearance but also restore functional autonomy.

References are important to the reader; therefore, each citation must be complete and correct. If at all possible, references should be commonly available publications.

REFERENCES

[1] Ivan Vujaklija, Dario Farina, and Oskar C Aszmann. “New developments in prosthetic arm systems”. In: *Orthopedic research and reviews* (2016), pp. 31–39.

[2] Sharlene N Flesher et al. “A brain-computer interface that evokes tactile sensations improves robotic arm control”. In: *Science* 372.6544 (2021), pp. 831–836.

[3] Bradley J Edelman et al. “Non-invasive brain-computer interfaces: state of the art and trends”. In: *IEEE reviews in biomedical engineering* (2024).

[4] Xiaolong Wu et al. “A review of motor brain-computer interfaces using intracranial electroencephalography based on surface electrodes and depth electrodes”. In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 32 (2024), pp. 2408–2431.

[5] Hongguang Pan et al. “Review of closed-loop brain–machine interface systems from a control perspective”. In: *IEEE Transactions on Human-Machine Systems* 52.5 (2022), pp. 877–893.

[6] Arjan Gijsberts et al. “Movement error rate for evaluation of machine learning methods for sEMG-based hand movement classification”. In: *IEEE transactions on neural systems and rehabilitation engineering* 22.4 (2014), pp. 735–744.

[7] Manfredo Atzori et al. “Characterization of a benchmark database for myoelectric movement classification”. In: *IEEE transactions on neural systems and rehabilitation engineering* 23.1 (2014), pp. 73–83.

[8] Manfredo Atzori and Henning Müller. "The Ninapro database: a resource for sEMG naturally controlled robotic hand prosthetics". In: *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE. 2015, pp. 7151–7154.

[9] Sasha Salter et al. "emg2pose: A large and diverse benchmark for surface electromyographic hand pose estimation". In: *Advances in Neural Information Processing Systems* 37 (2024), pp. 55703–55728.

[10] Tony Shu et al. "Tissue-integrated bionic knee restores versatile legged movement after amputation". In: *Science* 389.6756 (2025), eadv3223.

[11] Ziming Chen et al. "Tactile Sensing Enables Shared Control of Prosthetic Hand with Multi-Stage Grasping and Force Level Switching Functions". In: *Tsinghua Science and Technology* (2025).

[12] Sriramana Sankar et al. "A natural biomimetic prosthetic hand with neuromorphic tactile sensing for precise and compliant grasping". In: *Science Advances* 11.10 (2025), eadr9300.

[13] Patricia Capsi-Morales et al. "Merging motoneuron and postural synergies in prosthetic hand design for natural bionic interfacing". In: *Science Robotics* 10.98 (2025), eado9509.

[14] Yufei Ye et al. "Affordance diffusion: Synthesizing hand-object interactions". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023, pp. 22479–22489.

[15] Robin Rombach et al. "High-resolution image synthesis with latent diffusion models". In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022, pp. 10684–10695.

[16] Chenfei Wu et al. "Qwen-image technical report". In: *arXiv preprint arXiv:2508.02324* (2025).

[17] Team Wan et al. "Wan: Open and advanced large-scale video generative models". In: *arXiv preprint arXiv:2503.20314* (2025).

[18] Yi-Lin Wei et al. "OmniDexGrasp: Generalizable Dexterous Grasping via Foundation Model and Force Feedback". In: *arXiv preprint arXiv:2510.23119* (2025).

[19] Junbang Liang et al. "Dreamitate: Real-world visuomotor policy learning via video generation". In: *arXiv preprint arXiv:2406.16862* (2024).

[20] Joel Jang et al. "DreamGen: Unlocking Generalization in Robot Learning through Video World Models". In: *arXiv preprint arXiv:2505.12705* (2025).

[21] Shivansh Patel et al. "Robotic Manipulation by Imitating Generated Videos Without Physical Demonstrations". In: *arXiv preprint arXiv:2507.00990* (2025).

[22] Qingwen Bu et al. "Closed-loop visuomotor control with generative expectation for robotic manipulation". In: *Advances in Neural Information Processing Systems* 37 (2024), pp. 139002–139029.

[23] Ying Li et al. “ManipDreamer: Boosting Robotic Manipulation World Model with Action Tree and Visual Guidance”. In: *arXiv preprint arXiv:2504.16464* (2025).
[24] Xuhui Kang and Yen-Ling Kuo. “Incorporating task progress knowledge for subgoal generation in robotic manipulation through image edits”. In: *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE. 2025, pp. 7490–7499.
[25] Mengda Xu et al. “DexUMI: Using Human Hand as the Universal Manipulation Interface for Dexterous Manipulation”. In: *arXiv preprint arXiv:2505.21864* (2025).
[26] Yufei Wang et al. “Robogen: Towards unleashing infinite data for automated robot learning via generative simulation”. In: *arXiv preprint arXiv:2311.01455* (2023).
[27] Minjie Cai, Kris M Kitani, and Yoichi Sato. “Understanding Hand-Object Manipulation with Grasp Types and Object Attributes.” In: *Robotics: science and systems*. Vol. 3. 2016.